Fibonacci and Catalan Numbers

Hong Kong IMO Training

1 Must Knows

The following things are all well-known in the olympiad community. There should be no problem in citing them without proof on a competition. If you do not know these results, you should work out the proofs. Hints are provided to guide you.

1.1 Fibonacci Numbers

Define the Fibonacci numbers by $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$.

Proposition 1.1 (Binet)
For all n,
$$F_n = \frac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{2^n\sqrt{5}}.$$

This can be proven easily by straightforward induction. However, this does not provide insight in how one would find the formula independently. For that purpose, we provide two outlines of proofs that enable you to solve general linear recurrences.

Proof. The first proof uses the characteristic polynomial of a recurrence relation. This method can be extended to solve most linear recurrences.

(a) Call a sequence (a_n) *Fibonacci-like* if it satisfies the recurrence $a_{n+1} = a_n + a_{n-1}$. Show that if two sequences (a_n) and (b_n) are Fibonacci-like, then the sequence $(a_n + b_n)$ is also Fibonacci-like.

(b) Let α and β be the roots of the polynomial $x^2 - x - 1 = 0$. Show that the sequences (α^n) and (β^n) are Fibonacci-like.

(c) Use the initial values of F_n to find real numbers λ_1 and λ_2 such that
$$F_n = \lambda_1\alpha^n + \lambda_2\beta^n.$$

(d) Conclude.

□

Proof. This proof uses generating functions, which can be used to solve a wide variety of problems, including more complex recursions.

(a) Let $F(X) = \sum_{n\geq 0} F_n X^n$ and solve for $F(X)$.

(b) Let α and β be the roots of $x^2 - x - 1$. Find constants A and B such that
$$F(X) = \frac{A}{1-\alpha X} + \frac{B}{1-\beta X}.$$

(c) Show that $\frac{1}{1-rX} = \sum_{n\geq 0} r^n X^n$.

(d) Conclude.

□

Proposition 1.2 (Periodicity)
The Fibonacci sequence is periodic modulo m for any positive integer m.

(a) Show that if there are two pairs (F_a, F_{a+1}) and (F_b, F_{b+1}) which are congruent mod m, then the sequence is eventually periodic. Use this to prove that (F_n) is eventually periodic mod m.

(b) Use the recurrence relation $a_{n-1} = a_{n+1} - a_n$ to show that the period starts at F_0.

Proposition 1.3 (Zeckendorf)
Every positive integer can be represented uniquely as the sum of one or more distinct Fibonacci numbers in a sum that does not include two consecutive Fibonacci numbers.

(a) Prove the existence of a Zeckdendorf representation using induction.

(b) Suppose that $F_m \leq n < F_{m+1}$. Prove that the Zeckendorf representation must include F_m. Use this to conclude uniqueness.

1.2 Catalan Numbers

The Catlan numbers are defined $C_0 = 1$ and $C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$ for $n \geq 0$.

Proposition 1.4 (Common appearances of Catalan numbers)
Each of the following are equal to C_n.

(a) The number of paths from a corner to the opposite corner along the grid lines of an $n \times n$ grid which use exactly $2n$ moves and do not cross the diagonal of the square.

(b) The number of expressions containing n pairs of parentheses which are correctly matched.

(c) The number of ways of draw n line segments among $2n$ points on a circle such that no two segments intersect (even at an endpoint).

(d) The number of triangulations of a convex polygon with $n + 2$ vertices.

(e) The number of full binary trees with $n + 1$ leaves. (A full binary tree is a graph that is a tree where every node has either 0 or 2 children.)

(f) The number of ways to place the numbers $1, 2, \ldots, 2n$ in a $2 \times n$ grid such that each row and column is increasing.

The proof is left as an exercise.

Proposition 1.5
For all n,

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Proof. (a) Label the grid such that the top right corner is (n, n). Find a point (a, b) such that there is a bijection between the number of paths to (a, b) and paths that cross the diagonal.

(b) Conclude a formula for C_n.

□

Proof. (a) Let $C(X) = \sum_{n\geq 0} C_n X^n$. Show that

$$C(X) = \frac{1-\sqrt{1-4X}}{X}.$$

(b) Apply the binomial theorem to $\sqrt{1-4X}$ to solve for C_n.

□

2 Good to Knows

The following are properties of Fibonacci numbers that occasionally show up in problems. These can be useful to know and can most likely be cited without proof. However, you most likely will not remember all of these results and may need to refind them on your own during a contest.

(a) For all n,

$$F_1 + F_2 + \cdots + F_n = F_{n+2} - 1.$$

(b) For all n.

$$F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}.$$

(c) For all n and r,

$$F_n^2 - F_{n+r}F_{n-r} = (-1)^{n-r}F_r^2.$$

(d) For all $m > n$,

$$F_m F_{n+1} - F_{m+1}F_n = (-1)^n F_{m-n}.$$

(e) If $m \mid n$, then $F_m \mid F_n$.

(f) For all m and n,

$$\gcd(F_m, F_n) = F_{\gcd(m,n)}.$$

(g) Define the *Lucas numbers* L_n by $L_1 = 1$, $L_2 = 3$ and $L_{n+1} = L_n + L_{n-1}$ for $n \geq 1$. Show that $F_{n-1} + F_{n+1} = L_n$.

3 Problems

3.1 Combinatorics

Problem 3.1 (Bangladesh 2020)**.** Urmi is playing a game on a computer. If the computer screen displays the number x, then in the next move, Urmi can do one of the following:

1. Replace x by $4x + 1$.

2. Replace x by the largest integer not greater than $x/2$.

Initially the computer screen displays 0. How many different integers less than or equal to 2020 can Urmi achieve through a sequence of moves? It is permitted for the number displayed on the screen to exceed 2020 during the sequence.

Problem 3.2 (USA TSTST 2019)**.** Let $\mathcal{S}$ be a set of 16 points in the plane, no three collinear. Let $\chi(\mathcal{S})$ denote the number of ways to draw 8 lines with endpoints in $\mathcal{S}$, such that no two drawn segments intersect, even at endpoints. Find the smallest possible value of $\chi(\mathcal{S})$ across all such $\mathcal{S}$.

Problem 3.3 (RMM 2018)**.** Let n be positive integer and fix $2n$ distinct points on a circle. Determine the number of ways to connect the points with n arrows (oriented line segments) such that all of the following conditions hold: each of the $2n$ points is a startpoint or endpoint of an arrow; no two arrows intersect; and there are no two arrows $\overrightarrow{AB}$ and $\overrightarrow{CD}$ such that A, B, C and D appear in clockwise order around the circle (not necessarily consecutively).

Problem 3.4 (China 2020)**.** Consider a triangulation of a convex polygon with 20 vertices. A subset of 10 of the 37 line segments in a triangulation is called a *perfect matching* if covers all 20 vertices. Among all possible triangulations, find the maximum number of perfect matchings.

Problem 3.5 (Putnam 2020)**.** Let a_n be the number of sets S of positive integers for which

$$\sum_{k\in S} F_k = n.$$

Find the largest number n such that $a_n = 2020$.

Problem 3.6 (Indonesia TST 2020)**.** Given a natural number n, define B_n as the set of all sequences $b_1, b_2, \ldots, b_n$ of length n such that $b_1 = 1$ and for every $i = 1, 2, \ldots, n$, then we have

$$b_{i+1} - b_i \in \{1, -1, -3, -5, -7, \ldots\}$$

Where $b_i > 0$ for all i. Find a closed form of $|B_n|$

Problem 3.7 (Estonia, USA)**.** A sequence of positive integers $(a_n)_{n\geq 1}$ is of Fibonacci type if it satisfies the recursive relation $a_{n+2} = a_{n+1} + a_n$ for all $n \geq 1$. Is it possible to partition the set of positive integers into an infinite number of Fibonacci type sequences?

3.2 Number Theory

Problem 3.8. Prove the properties listed in Section 2.

Problem 3.9. Let k be an integer such that $p = 5k + 1$ is a prime. Show that the period of the Fibonacci sequence modulo p divides $p - 1$.

Problem 3.10 (ELMO 2020)**.** Let k be a positive integer. Suppose that for every positive integer m there exists a positive integer n such that $m \mid F_n - k$. Must k be a Fibonacci number?

Problem 3.11 (Harvard-MIT Mathematics Tournament 2017)**.** What is the period of the Fibonacci sequence modulo 127?

Problem 3.12 (USA TSTST 2019)**.** Suppose P is a polynomial with integer coefficients such that for every positive integer n, the sum of the decimal digits of $|P(n)|$ is not a Fibonacci number. Must P be constant?